

16 Inheritance

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Meiosis produces haploid gametes and variation by crossing over and independent assortment. At fertilisation diploid number is restored. For independently assorting genes, multiply probabilities; for linkage, recombinant classes are less frequent than predicted by independent assortment. A chi-square test compares observed and expected counts but does not prove the null hypothesis true.

Worked example: follow the reasoning

For $Aa \times Aa$, probability of aa is $\frac{1}{4}$; in two independent offspring probability both aa is $(\frac{1}{4})^2 = \frac{1}{16}$.

Quick check · multiple choice

A What can crossing over during meiosis produce? [1]

- A Identical clones only
- B Doubled chromosome number always
- C Loss of all genes
- D New allele combinations

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Two heterozygous parents produce 80 offspring; 18 show a recessive phenotype. [3]
Calculate expected number under a 3:1 ratio and give one cause of deviation.

Final answer: _____

Answers and explanation

Quick check: D. New allele combinations

Question 1 [3 marks]

Expected recessive = $80/4 = 20$ [1]; observed 18 can differ by sampling chance [1]; mutation/selection or flawed model are other acceptable reasons [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.